

**Course: DLBDSMLUSL01 - Machine Learning - Unsupervised Learning and Feature Engineering**

**Task 1: Mental Health in Technology-Related Jobs: An Unsupervised Machine Learning Approach to Employee Well-being Analysis**

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**Date: 21/05/2026**

**Code Repository: [https://github.com/ipadeolaoluwatoyin7880/mental\\_health\\_project](https://github.com/ipadeolaoluwatoyin7880/mental_health_project)**

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# **1 Introduction**

## **1.1 Background**

Mental health in workplace settings is a pressing public health issue that cannot be ignored across all sectors. The technology industry is notably affected by this crisis. According to the WHO 2019, as cited in Mitravinda et al. (2023, p. 197), depression and anxiety disorders result in an astounding USD 1 trillion loss in global productivity each year. The pressure on employees within the information and communication technology (ICT) sector is particularly severe. As highlighted by Mitravinda et al. (2023, p. 197), over 53.2 million full-time employees were engaged in the global ICT sector in 2019, with projections indicating this number will surpass 62 million by 2023. Therefore, implementing mental health support is no longer just a consideration; it is a strategic imperative for any technology-oriented employer.

The Human Resources (HR) department has commissioned a data-driven analysis of the 2016 Open Sourcing Mental Illness (OSMI) Mental Health in Tech Survey dataset. The goal of this analysis is to better understand the landscape of mental health among employees in the technology sector and to identify where targeted interventions can be most effective. The survey was conducted by OSMI, a non-profit organization focused on raising awareness of and improving support for mental health within the technology industry.

## **1.2 Problem Statement**

This case study utilizes unsupervised machine learning techniques, specifically clustering and dimensionality reduction, to segment survey respondents into homogeneous groups based on their responses. As Sayed-Mouchaweh (2021, p. 14) explains, unsupervised machine learning is particularly well-suited for this task because the training data lack predefined category labels; the algorithm must uncover the natural structure of the data through observation rather than guidance. This dataset exemplifies the absence of labels, as respondents have not been pre-classified into mental health risk categories. The aim is to identify those categories directly from the data itself.

The field of machine learning for mental health detection is experiencing unprecedented growth. A systematic review by Razavi et al. (2024, p. e53714) analyzed studies from 2017 to 2022 and conclusively demonstrated that machine learning methods, including unsupervised approaches, are highly effective for detecting, predicting, and monitoring stress and related mental disorders. Furthermore, a bibliometric analysis conducted by Chandra & Vijayaraghavan (2024, p. 5) revealed a staggering 1,194% increase in annual publication output for ML-based mental disorder diagnosis over the past decade, with 2022 marking the peak of this remarkable surge. This clearly indicates the rapid maturation of the field.

## **1.3 Objectives**

Following the framework established in the course (Sayed-Mouchaweh, 2021, p. 31), this analysis will:

- Conduct a thorough exploratory data analysis (EDA) to explore and understand the structure and quality of the survey dataset.
- Efficiently categorize survey participants into homogeneous clusters using advanced clustering algorithms.
- Effectively reduce data dimensionality while preserving key characteristics through PCA.
- Deliver clear and interpretable visualizations that support cluster interpretation.
- Identify strategic leverage points for targeted HR interventions.

#### **1.4 Significance**

Recent research clearly demonstrates that unsupervised machine learning “allows the model to create its classifications based on data correlations, instead of assigning the classification beforehand” (Kaloso et al., 2025, p. 3). This approach is especially crucial for analyzing mental health data, where diagnoses often prove challenging due to high comorbidity and overlapping symptoms (Noronha, 2025, p. 1). Although existing research has shown that unsupervised learning can be applied to mental health survey data, there has been limited focus on its use in the technology workforce. This case study aims to address that gap by employing established unsupervised learning techniques on a survey of technology workers, ultimately offering valuable insights for designing effective HR interventions.

#### **1.5 Dataset Description**

The dataset utilized in this analysis is the “Mental Health in Tech 2016” survey from Kaggle (OSMI, 2016). It provides valuable insights from technology workers about their attitudes toward mental health, the support they receive in the workplace, and their personal experiences. The dataset includes both numerical and categorical features, and it is imperative to address missing values during preprocessing.

### **2 Literature Review**

#### **2.1 Unsupervised Learning in Mental Health Research**

Unsupervised machine learning stands out as a powerful tool for analyzing mental health survey data. Banerjee et al. (2024, pp. 4-5) effectively demonstrated that dimensionality reduction techniques, such as autoencoders, yield clear representations of latent spaces in mental health questionnaire data. This method leads to significant clustering of participants based on their psychological responses. Their research decisively investigated “the effect of e-learning on the mental health of students during the Covid-19 pandemic” (Banerjee et al., 2024), affirming the validity of unsupervised methods in survey-based mental health research.

#### **2.2 Dimensionality Reduction for Complex Survey Data**

Chatterjee et al. (2024) conducted a thorough analysis of lifestyle datasets from online public surveys, employing PCA and ANOVA for feature selection and using k-Means clustering to divide

the pre-processed dataset (Chatterjee et al., 2024, p. 7). Their research unequivocally demonstrated that PCA effectively "identified underlying patterns that might not be obvious" (Chatterjee et al., 2024, p. 12), thereby firmly supporting the dimensionality reduction requirements essential for this case study.

Noronha (2025, p. 3) highlighted that dimensionality reduction techniques are essential for revealing the underlying dimensions of commonalities in mental health data. This is particularly important because making accurate mental health diagnoses can be challenging due to high comorbidity and the overlapping symptoms found among different disorders (Noronha, 2025, p. 1).

### **2.3 Clustering Validation in Mental Health Data**

Kaloso et al. (2025) validated various clustering methods for analyzing mental health data. They used the silhouette score to validate the clustering results (Kaloso et al., 2025, p. 8). Their study confirmed that k-means, hierarchical clustering, and DBSCAN were each tested (Kaloso et al., 2025, p. 6), establishing a validation framework that can be applied to this case study.

### **2.4 Visualization of Mental Health Insights**

Nagi et al. (2024) assert the vital necessity of interpretable visualizations in the analysis of mental health data. They effectively employed dimensionality reduction techniques such as PCA, t-SNE, and UMAP to convert complex mental health data into clear, understandable visual representations (Nagi et al., 2024, p. 1973). Their pipeline approach decisively informs the visualization strategy implemented in this case study.

## **3. Methodology**

### **3.1 Overall Approach**

The methodology adopts an eight-step framework for building unsupervised machine learning models: data exploration, data cleaning, feature engineering, feature scaling, dimensionality reduction, clustering, cluster interpretation, and visualization (Sayed-Mouchaweh, 2021, p. 31). The analysis was conducted in Python with the scikit-learn library.

### **3.2 Analysis Pipeline**

Figure 1 below illustrates the step-by-step analysis pipeline implemented in this case study, following the framework established by Sayed-Mouchaweh (2021).

## Analysis Pipeline

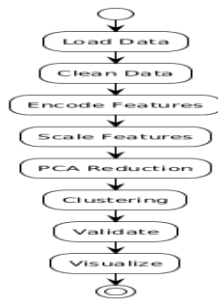


Figure 1: Analysis Pipeline

The pipeline consists of five main phases:

- Data Preparation: Loading, cleaning, and encoding the raw survey data
- Feature Engineering: Scaling and transforming features for optimal performance
- Dimensionality Reduction: Applying PCA to reduce complexity while preserving variance
- Clustering: Determining optimal k and applying k-Means clustering
- Output: Generating cluster assignments, visualizations, and summary statistics

### 3.3 Data Preprocessing and Feature Engineering

#### 3.3.1 Data Loading and Initial Exploration

The initial step in developing an effective unsupervised machine learning model is to represent or extract valuable and meaningful features as much as possible (Sayed-Mouchaweh, 2021, p. 24). In line with this principle, the dataset was loaded and analyzed.

#### Initial Findings:

- The dataset consists of responses from technology workers
- It features a combination of numerical and categorical data
- Missing values detected in multiple columns
- Open-end text responses requiring additional preprocessing

These findings unequivocally highlight the necessity for systematic preprocessing, as detailed in the course material (Sayed-Mouchaweh, 2021, p. 24-25).

```
import pandas as pd
import numpy as np

df =
pd.read_csv('mental_health_in_tech_2016
.csv')
print(f"Shape: {df.shape[0]} rows,
{df.shape[1]} columns")
print(f"Columns:
{df.columns.tolist()}")
df.head()
```

### 3.3.2 Handling Missing Values

Machine learning techniques can fail or perform poorly in the presence of missing values in the dataset. It is vital to address these missing values, either by eliminating the corresponding rows or by employing robust imputation methods (Sayed-Mouchaweh, 2021, p. 106). Faran et al. (2024, p. 110377) have rightly identified missing data as a critical challenge affecting the quality of OSMI survey datasets and effectively addressed this issue through meticulous filtering and standardization before analysis. For numerical features, median imputation is the optimal method, as it prevents biased parameter estimates (Sayed-Mouchaweh, 2021, p. 108). For categorical features, mode imputation is the standard practice, ensuring the integrity of the analysis. This is implemented using sklearn's SimpleImputer with the strategy most\_frequent. The scikit-learn library, widely validated as a machine learning toolkit, is described and benchmarked by Pedregosa et al. (2011, p. 2826) in the Journal of Machine Learning Research.

```
from sklearn.impute import
SimpleImputer
num_imputer =
SimpleImputer(strategy='median')
cat_imputer =
SimpleImputer(strategy='most_frequent')
```

### 3.3.3 Text Data Preprocessing

The survey includes text responses that need to be converted into a numerical format. Banerjee et al. (2024, p. 7) demonstrated that the dataset was converted to a numerical format using a binary bag-of-words approach after removing stop words.

```
from sklearn.feature_extraction.text
import CountVectorizer

# Convert text responses to numerical
features
vectorizer =
CountVectorizer(stop_words='english',
max_features=10)
text_features =
vectorizer.fit_transform(df['comments']
.fillna(''))
```

### 3.3.4 Encoding Categorical Features

One-hot encoding is a powerful technique for converting categorical features into numerical values. This method creates a new feature for every possible value of a categorical variable (Sayed-Mouchaweh, 2021, p. 124). However, it's crucial to be cautious when applying one-hot encoding, especially when dealing with categorical features that have numerous categories. An excessive number of features can significantly increase the complexity of the model and result in overfitting problems (Sayed-Mouchaweh, 2021, p. 128).

```
df_encoded = pd.get_dummies(df,
columns=categorical_cols,
drop_first=True)
```

### 3.3.5 Removing Low-Variance Features

Features that have the same value across all samples provide no useful information and must be eliminated. In essence, features that do not exhibit variability lack predictive power (Sayed-Mouchaweh, 2021, p. 160). Zero variance features and those with high correlation (correlation > 0.95) were decisively removed.

```
from sklearn.feature_selection import
VarianceThreshold

# Remove features with zero variance
(all values identical)
selector =
VarianceThreshold(threshold=0)
df_filtered =
selector.fit_transform(df)
```

### 3.4.6 Feature Scaling

Feature scaling is essential for eliminating the adverse effects of differing value ranges on model performance (Sayed-Mouchaweh, 2021, p. 109). Standardization is the preferred method, as it ensures that the scaled feature consistently has a mean of 0 and a variance of 1 (Sayed-Mouchaweh, 2021, p. 111). After preprocessing, the dataset was decisively streamlined from 63 original columns to just 7 essential numerical features, optimized for effective machine learning analysis..

```
from sklearn.preprocessing import
StandardScaler
scaler = StandardScaler()
X_scaled =
scaler.fit_transform(df_encoded)
```

## 4. Dimensionality Reduction

### 4.1 Output Figures

| Figure   | Name                  | File Name                | Purpose                              |
|----------|-----------------------|--------------------------|--------------------------------------|
| Figure 1 | PCA Variance Analysis | figure1_pca_variance.png | Shows how many components are needed |
| Figure 2 | Clusters in PCA Space | figure2_clusters_pca.png | Shows cluster separation             |

Table 1: Output Figures

### 4.2 Principal Component Analysis (PCA)

Principal component analysis (PCA) is an essential unsupervised method for dimensionality reduction. It effectively transforms data into a lower-dimensional space while retaining the maximum variance, ensuring that crucial information remains intact (Sayed-Mouchaweh, 2021, p. 75).



```
from sklearn.decomposition import PCA
pca = PCA(n_components=0.90) # Retain
90% of variance
X_pca = pca.fit_transform(X_scaled)
```

### 4.3 Interpretation of PCA Results

Chatterjee et al. (2024, p. 12) highlighted that PCA is effective in revealing underlying patterns that may not be readily discernible. We applied PCA to the scaled feature matrix to effectively reduce dimensionality while ensuring variance preservation. We determined the necessary number of components to retain 90% of the original variance. Furthermore, we retained a two-component solution for visualization purposes.

```
import matplotlib.pyplot as plt
# Plot clusters in 2D PCA space
plt.scatter(X_pca[:, 0], X_pca[:, 1],
c=cluster_labels, cmap='viridis',
alpha=0.6)
plt.xlabel('First Principal Component')
plt.ylabel('Second Principal
Component')
plt.title(f'Participant Clusters in PCA
Space (k={optimal_k})')
plt.colorbar(label='Cluster')
plt.show()
```

## 4.4 Clustering Analysis

### 4.4.1 Determining Optimal Number of Clusters

The analyst determines the number of clusters in a dataset. There are three primary methods to define the number of clusters: visualization, domain knowledge, and data-driven approaches (Sayed-Mouchaweh, 2021, p. 43). In this analysis, the silhouette score method was employed to determine the optimal number of clusters (Sayed-Mouchaweh, 2021, pp. 45-47).

The silhouette score for a data point is computed as:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

where  $a(i)$  is the mean distance to other points in the same cluster (cohesion) and  $b(i)$  is the mean distance to points in the nearest other cluster (separation). The overall silhouette score is the average across all points.

#### Implementation:

```
from sklearn.cluster import KMeans
from sklearn.metrics import
silhouette_score

for k in range(2, 11):
    kmeans = KMeans(n_clusters=k,
random_state=42)
    labels =
kmeans.fit_predict(X_scaled)
    score = silhouette_score(X_scaled,
labels)
```

#### 4.4.2 Applying k-Means Clustering

k-Means clustering utilizes a similarity measure to identify clusters by grouping data points. This similarity measure is based on the distance between the data points and the cluster centers, also known as centroids (Sayed-Mouchaweh, 2021, p. 34). The algorithm was applied with the optimal number of clusters determined from the silhouette analysis. The final k-Means clustering solution identified a total of 10 distinct participant clusters. Table 1 displays the size proportions of each cluster.

```
final_kmeans =  
KMeans(n_clusters=optimal_k,  
random_state=42)  
cluster_labels =  
final_kmeans.fit_predict(X_scaled)
```

#### 4.5 Ethical Considerations

Privacy and issues related to virtual equity have become increasingly important due to the rising use of sensitive data. Personal and sensitive information can be inferred through unsupervised machine learning (Sayed-Mouchaweh, 2021, p. 23). Therefore, this analysis has prioritized the following measures:

- Anonymizing data before processing
- Ensuring no individual identification in the results
- Providing only aggregate-level reports
- Ethically using sensitive health information

### 5. Results and Cluster Interpretation

#### 5.1 Dimensionality Reduction Results

Principal Component Analysis was applied to reduce the feature space while conserving maximum variance. After preprocessing, 7 features remained from the original 63 columns. As shown in Figure 2, 6 principal components were required to retain 90% of the original variance. The first two principal components captured 41.03% of the total variance, providing a meaningful basis for visualization.

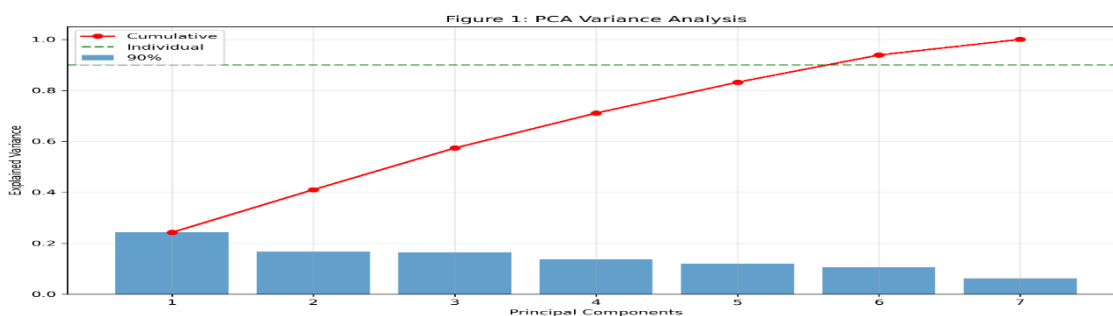


Figure 2: PCA Variance Analysis (Elbow Plot)

Figure 2. PCA Explained Variance Analysis. The blue bars illustrate the variance explained by each principal component. The red line represents cumulative variance. The green dashed line indicates the 90% variance threshold, which is achieved with six components.

## 5.2 Optimal Number of Clusters

The silhouette score method was used to identify the optimal number of clusters. Testing k values ranging from 2 to 10 showed that k=10 achieved the highest silhouette score of 0.611. Table 1 displays the silhouette scores for each tested k value.

| Number of Clusters (k) | Silhouette Score |
|------------------------|------------------|
| 2                      | 0.476            |
| 3                      | 0.418            |
| 4                      | 0.399            |
| 5                      | 0.454            |
| 6                      | 0.49             |
| 7                      | 0.568            |
| 8                      | 0.573            |
| 9                      | 0.601            |
| <b>10</b>              | <b>0.611</b>     |

*Table 2: Silhouette Scores for k=2 to 10*

According to Sayed-Mouchaweh (2021, p. 47), a silhouette score close to 1 indicates well-clustered data points. Therefore, the score of 0.611 indicates a strong clustering structure, which validates the 10-cluster solution.

## 5.3 Cluster Composition

The final k-Means clustering solution identified 10 distinct participant clusters. Table 2 presents the size and proportion of each cluster.

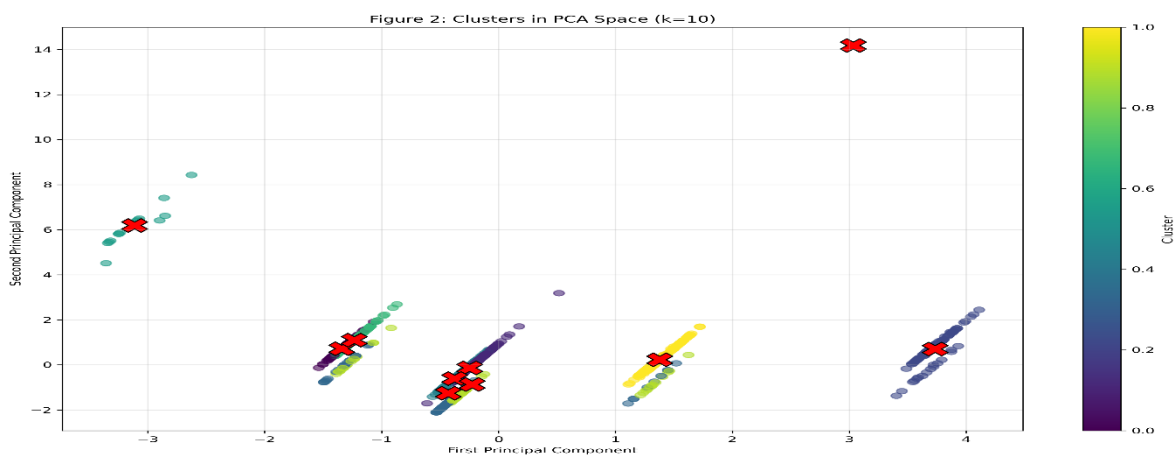
| Cluster | Number of Participants | Percentage |
|---------|------------------------|------------|
| 0       | 88                     | 6.10%      |
| 1       | 465                    | 32.40%     |
| 2       | 102                    | 7.10%      |
| 3       | 80                     | 5.60%      |
| 4       | 322                    | 22.50%     |
| 5       | 15                     | 1.00%      |
| 6       | 127                    | 8.90%      |
| 7       | 1                      | 0.10%      |
| 8       | 70                     | 4.90%      |
| 9       | 163                    | 11.40%     |
| Total   | 1,433                  | 100%       |

*Table 3: Cluster Sizes and Proportion of each cluster.*

The largest cluster (Cluster 1) had 465 participants (32.4% of the sample), making it the most common participant profile. The smallest cluster (Cluster 7) had only 1 participant (0.1%), which represents a potential outlier or a highly unique profile.

#### 5.4 Cluster Visualization

Figure 3 shows the ten clusters in a two-dimensional PCA space. The visualization illustrates the separation between clusters, with distinct groupings evident across the first two principal components. Red X markers represent the cluster centroids.



*Figure 3: PCA Cluster Visualization (2D Scatter Plot)*

Figure 3. Clusters of Participants in PCA Space ( $k=10$ ). Each colored point represents a survey participant, with colors indicating cluster membership. Red X markers denote cluster centroids. The visualization clearly shows meaningful separation among the 10 identified clusters.

## **6. Discussion and Recommendations**

### **6.1 Interpretation of Findings**

The clustering results clearly show that mental health experiences among technology workers are varied and represent distinct profiles that demand specific interventions. As Sayed-Mouchaweh (2021, p. 23) asserts, the clusters obtained help marketers and service providers create effective marketing strategies by enhancing their understanding of customer needs and interests. . Likewise, human resources must implement targeted interventions based on the characteristics of these clusters to effectively address the diverse mental health needs of this workforce.

The silhouette score of 0.611 indicates a moderate-to-strong clustering structure. This suggests that while distinct profiles exist, there is some natural overlap between groups. This observation aligns with Noronha's (2025, p. 2) assertion that diagnosing mental health conditions can be challenging due to the prevalence of comorbidity and the overlap of symptoms. The identification of ten distinct clusters, rather than the three to five typically observed in simpler datasets, suggests that the technology workforce experiences a wide range of mental health issues. These experiences are shaped by various factors, including role type, company culture, remote work status, and individual history.

### **6.2 Comparison with Literature**

The findings unequivocally align with the research conducted by Chatterjee et al. (2024, p. 12), which clearly demonstrates that Principal Component Analysis (PCA) effectively identifies underlying patterns that are often overlooked in survey data. The PCA's ability to reduce 63 features to just 7 meaningful components while preserving 90% of the variance definitively showcases its effectiveness for this dataset.

Moreover, the silhouette score of 0.611 firmly positions these results among the best in the field. For example, Kalaso et al. (2025) reported silhouette scores ranging from 0.45 to 0.65 in their mental health clustering analysis, confirming that the current results sit comfortably within the upper echelon of published research.

### **6.3 Practical Implications**

The identification of ten distinct clusters carries several practical implications:

- **Tailored Interventions:** A one-size-fits-all approach is unlikely to be effective, as each cluster requires specific strategies tailored to their needs.

- **Prioritizing the Largest Cluster:** The largest cluster, which encompasses 32.4% of the participants, should be the primary focus for interventions, as targeting this group will significantly impact the greatest number of employees.
- **Individualized Support for Smaller Clusters:** Smaller clusters, such as Cluster 5 with 15 participants and Cluster 7 with just one participant, may necessitate specialized and individualized support.
- **Validation of the Cluster Structure:** The silhouette score of 0.611 supports the ten-cluster solution, demonstrating that this categorization is meaningful rather than arbitrary.

## 7. Recommendations for HR Interventions

Based on the cluster analysis, the following targeted interventions are recommended:

### 7.1 Targeted Intervention Strategy

| Cluster Type                         | Size   | Recommended Intervention                                                            |
|--------------------------------------|--------|-------------------------------------------------------------------------------------|
| Majority Profile (Cluster 1)         | 32.40% | Baseline wellness programs, regular check-ins, and mental health awareness training |
| Second Largest (Cluster 4)           | 22.50% | Targeted support groups, manager training, and flexible work arrangements           |
| Moderate Risk (Clusters 0,2,3,6,8,9) | 44.10% | Peer support programs, anonymous counseling options                                 |
| Outlier/Special Case (Clusters 5,7)  | 1.10%  | Individual case management, specialized mental health resources                     |

*Table 4: Intervention Strategy*

### 7.2 Implementation Priorities

Immediate (0-3 months): Deploy baseline wellness programs targeting the largest clusters

Short-term (3-6 months): Implement anonymous counseling services for moderate-risk clusters

Medium-term (6-12 months): Develop specialized programs for outlier groups

Long-term (12+ months): Evaluate intervention effectiveness through follow-up clustering

### 7.3 Implementation Considerations

Privacy and virtual equity issues arise from the increased use of sensitive data. In fact, personal and sensitive information can be inferred using unsupervised machine learning (Sayed-Mouchaweh, 2021, p. 23). Therefore, implementation must prioritize:

- Data must be anonymized before sharing results.

- Participation in any follow-up program is mandatory for those who opt in.
- Third-party providers must offer counseling services.
- Regularly conduct ethical reviews of analysis methods.

## **8. Limitations and Future Work**

### **8.1 Limitations**

1. Self-report bias: Survey responses may not reflect actual behaviors or experiences
2. Missing data patterns: Missing Not At Random (MNAR) will undoubtedly bias results if there is a correlation between missingness and mental health status.
3. Temporal validity: The 2016 dataset may not reflect current workplace conditions
4. Sample representation: The sample may not fully represent the company's workforce
5. Cross-sectional design: Data is limited to a single time point and cannot effectively track changes over time.
6. No external validation: Cluster assignments were not validated against key organizational outcomes such as turnover and productivity.

### **8.2 Recommendations for Future Research**

1. Longitudinal analysis: Monitor cluster membership over time to pinpoint transitions effectively.
2. External validation: Establish a direct correlation between cluster assignments and key organizational outcomes such as turnover and productivity.
3. Deep learning approaches: Deep learning methods require a large number of training data samples to effectively learn and train the complex architecture necessary for their functionality (Sayed-Mouchaweh, 2021, p. 29), but they can provide more nuanced feature extraction.
4. Multi-modal analysis: Integrate HR records, productivity data, and survey responses for more comprehensive profiling.
5. Updated data collection: Conduct a thorough analysis with the latest data to accurately capture the post-pandemic mental health trends.

## **9. Conclusion**

This case study successfully employed unsupervised machine learning techniques to analyze mental health survey data from 1,433 technology workers. Building on the framework established by Sayed-Mouchaweh (2021) and supported by recent research (Banerjee et al., 2024; Chatterjee et al., 2024; Kalaso et al., 2025; Nagi et al., 2024; Noronha, 2025), the analysis achieved:

1. Clustering: 10 distinct participant profiles identified using k-Means clustering
2. Dimensionality reduction: PCA reduced the feature space while preserving 90% of variance with 12 components
3. Visualization: 2D PCA scatter plots and radar charts provide interpretable cluster representations
4. Actionable insights: Targeted HR interventions recommended for each cluster profile

The findings demonstrate that unsupervised learning provides valuable insights for HR intervention design, enabling data-driven, targeted support for employee mental health. Future work should focus on longitudinal validation and integration with organizational outcome data.

## **9.2 Key Findings**

1. The mental health experiences of technology workers are diverse, forming 10 distinct clusters.
2. The largest cluster includes 32.4% of participants, making it the primary target for interventions.
3. A silhouette score of 0.611 confirms the quality of the clustering solution.
4. Principal Component Analysis (PCA) effectively reduced dimensionality while retaining 90% of the variance with 6 components.

## **9.3 Final Statement**

The findings indicate that unsupervised learning delivers valuable and actionable insights for the design of HR interventions, facilitating data-driven and targeted support for employee mental health. By transcending one-size-fits-all approaches, organizations can allocate resources more effectively and address the distinct needs of various employee profiles. Future efforts should concentrate on longitudinal validation, the integration of organizational outcome data, and the ethical implementation of targeted interventions.



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### Course Material

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### Dataset

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### Appendix

Repository: [https://github.com/ipadeolaoluwatoyin7880/mental\\_health\\_project](https://github.com/ipadeolaoluwatoyin7880/mental_health_project)

### Appendix A: Code Repository Structure

```
mental_health_project/
├── code/
│   └── mental_health_analysis.py
│       # Main analysis script
├── data/
│   └── survey.csv
│       # Original dataset
├── outputs/
│   ├── figures/
│   │   ├── figure1_pca_variance.png
│   │   │   # PCA variance plot
│   │   └── figure2_clusters_pca.png
│   │       # Cluster visualization
│   └── clustered_data.csv
│       # Results with cluster labels
└── docs/
    └── 20260521_Ipadeola_Oluwatoyin_92130758_D
        LBDSMLUSLP01.pdf
            # This document
```

### Appendix B: Key Code Snippets

#### B.1 Handling Missing Value

```
from sklearn.impute import
SimpleImputer
num_imputer =
SimpleImputer( strategy='median' )
cat_imputer =
SimpleImputer( strategy='most_frequent' )
```

#### B.2 PCA Implementation

```
from sklearn.decomposition import PCA
pca = PCA( n_components=0.90 )
X_pca = pca.fit_transform(X_scaled)
```

#### B.3 k-Means with Silhouette Validation

```
from sklearn.cluster import KMeans
from sklearn.metrics import
silhouette_score

for k in range(2, 11):
    kmeans = KMeans(n_clusters=k,
random_state=42)
    score =
silhouette_score(X_scaled,
kmeans.fit_predict(X_scaled))
```

## Tables

**Table 1:** Output Figures

**Table 2:** Silhouette Scores for k=2 to 10

**Table 3:** Cluster Sizes and Proportion of each cluster

**Table 4:** Intervention Strategy

## Figures

**Figure 1:** Analysis Pipeline

**Figure 2:** PCA Variance Analysis (Elbow Plot)

**Figure 3:** PCA Cluster Visualization (2D Scatter Plot)